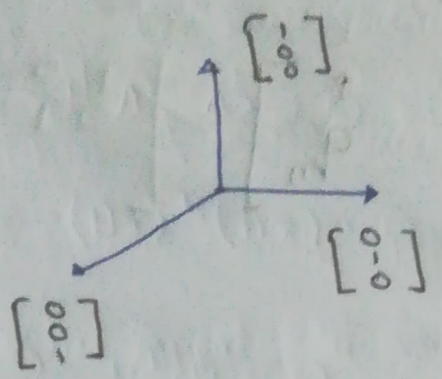
Lecture 13 (12/07/21)

Orthogonal matrices and Gram - Schimdt process

- "Columns are independent if they are perpendicular unit vectors" (Theorem)



Ortho-normal vectors (q_i)
- Orthogonal to each other
- Unit (value = 1)
  (normalised value)

Property

$$q_i^T \cdot q_j = \begin{cases} 0, & i \neq j \\ 1, & i = j \end{cases}$$

Orthonormal Matrix, $Q = \begin{bmatrix} \uparrow & \uparrow & \uparrow & & \uparrow \\ q_1 & q_2 & q_3 & \cdots & q_n \\ \downarrow & \downarrow & \downarrow & & \downarrow \end{bmatrix}$

If the orthonormal matrix is a square matrix, it is termed as → "Orthogonal Matrix"

$$Q^T = \begin{bmatrix} \leftarrow q_1 \rightarrow \\ \leftarrow q_2 \rightarrow \\ \leftarrow q_3 \rightarrow \\ \vdots \\ \leftarrow q_n \rightarrow \end{bmatrix}$$

Property of Orthogonal matrix

$$Q^T \cdot Q = \begin{bmatrix} \leftarrow q_1 \rightarrow \\ \vdots \\ \leftarrow q_n \rightarrow \end{bmatrix} \begin{bmatrix} \uparrow & & \uparrow \\ q_1 & \cdots & q_n \\ \downarrow & & \downarrow \end{bmatrix} = \begin{bmatrix} 1 & 0 & & 0 \\ 0 & 1 & & 0 \\ \vdots & & \ddots & \vdots \\ & & & 1 \end{bmatrix}$$

$\rightarrow$ $\boxed{Q^T \cdot Q = I}$

$\boxed{Q^T = Q^{-1}}$

All permutation matrices are orthogonal matrices

Example:

Permutation Matrix,

$$Q = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \qquad Q^T = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$Q^T \cdot Q = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

Prove orthonormal matrix → satisfy 2 properties

1) $Q = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ rotation matrix ✓

Orthogonal ✓

Values = 1, 1

dot product = 0

$= \sqrt{\cos^2\theta + \sin^2\theta} = 1$

2) $Q = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ Orthogonal ✗

Values = $\sqrt{2}$ $\sqrt{2}$ ✗

dot product = 0

3) $Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ Orthogonal ✓

Values = 1 1

dot product = 0

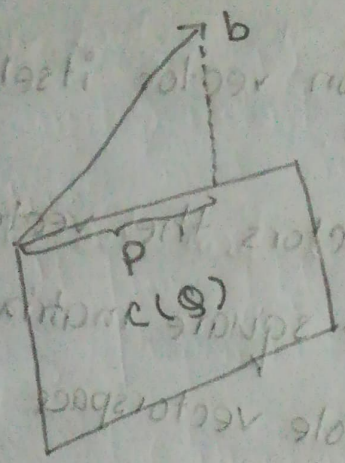
$$Q = \boxed{\frac{1}{2}} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$$ Orthogonal ✓

$$Q = \boxed{\frac{1}{3}} \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ -2 & 2 \end{bmatrix}$$ Orthonormal ✓

(rectan

Previously we saw that if theres a col^n space $C(A)$ of a matrix A, and there's a vector b ($\notin C(A)$), so we took its projection on $C(A)$.

So, what if we replace the matrix A with a orthogonal matrix Q



$Q \rightarrow$ orthogonal

$$Q = P = Q\,(\underbrace{Q^T . Q}_{I})^{-1}\,Q^T$$

$$= \boxed{Q\,Q^T}$$

if square matrix, $\boxed{Q . Q^T = Q^T . Q = I}$

$$Ax = b$$

$$\Rightarrow A^T . A . \hat{x} = A^T . b$$

$$\Rightarrow \underbrace{Q^T Q}_{I}\,\hat{x} = Q^T . b$$

$$\Rightarrow \boxed{\hat{x} = Q^T . b}$$

$$p = A\hat{x} = Q\hat{x} = \underbrace{Q\,Q^T}\,.\,b$$ for square matrix, $QQ^T = I$

$\therefore \boxed{p = b}$ for square matrix

$\boxed{p = b}$ Projection is the solution vector itself.

Incase of perpendicular unit vectors, the vectors are always independent, So if we take one square matrix and all vectors are independent, it spans the whole vectorspace ($C(A) = R^n$), If so, no b can be outside the columnspace, so we get the projection as the vector itself.

- If we have 3 vectors (not unit/orthogonal) but we want to make it orthogonal matrix

$[a, b, c] \rightarrow$ random but independent vector.

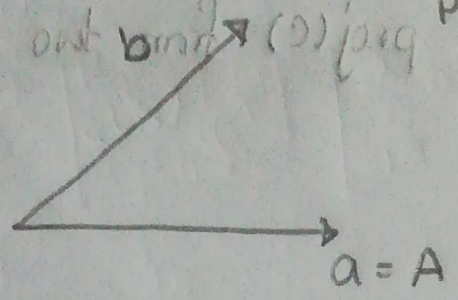
<u>Gram - Schimdt</u>

1. Construct 3 orthogonal vectors from a, b, c to A, B, C
2. From A, B, C, we normalize them

$$\frac{A}{\|A\|}, \frac{B}{\|B\|}, \frac{C}{\|C\|}$$

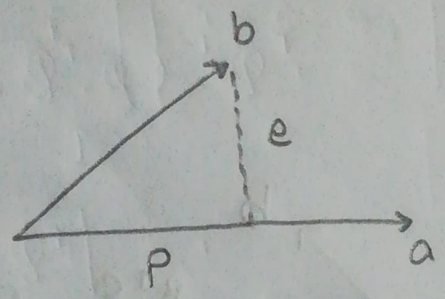
3. Arrange as $Q = \begin{bmatrix} \uparrow & \uparrow & & \uparrow \\ a_1 & a_2 & \cdots & a_3 \\ \downarrow & \downarrow & & \downarrow \end{bmatrix}$

Example:



pick one of the vectors as reference, here A

$e \perp A$



$b = p + e$

$p \perp e \qquad e \perp a$

$e = B$

$e = b - p = b - A\hat{x} = b - a \cdot \dfrac{a^T \cdot b}{a^T \cdot a}$

$= b - A \cdot \dfrac{(A^T \cdot b)}{A^T \cdot A}$

$\therefore \boxed{B = b - A \cdot \dfrac{A^T \cdot b}{A^T \cdot A}}$

$[\perp \text{ to } A]$

$A^T \cdot B = A^T \left(b - A \cdot \dfrac{A^T \cdot b}{A^T \cdot A} \right)$

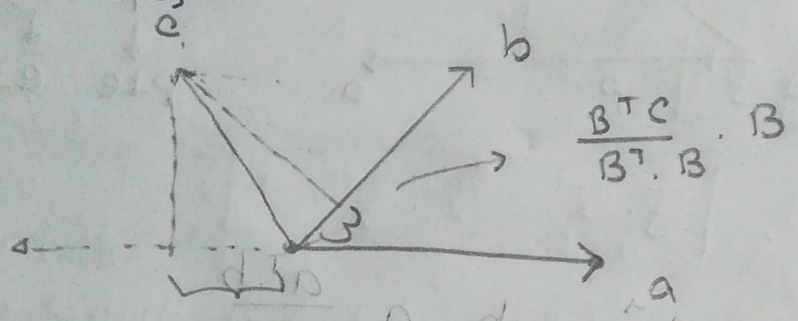
$= \dfrac{(A^T \cdot b) - (A^T A)(A^T b)}{A^T \cdot A} = 0$

$(A \perp B)$

We subtracted the projection from the actual vector and obtained error, where $e \perp A$.

For 3 vectors, cancel out proj(c) from two other vectors

For 3 vectors,



$$\frac{A^T . C}{A^T . A} . A$$

Check
$A . B = 0$
$B . C = 0$

$$C = c - \frac{A^T . C}{A^T . A} . A - \frac{B^T . C}{B^T B} . B \quad ✓$$

Last step: Make them normal.

$$\underbrace{\frac{A}{\| A \|}}_{q_1} , \underbrace{\frac{B}{\| B \|}}_{q_2} , \underbrace{\frac{C}{\| C \|}}_{q_3}$$

Exercise

$$a = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \quad b = \begin{bmatrix} 2 \\ 0 \\ -2 \end{bmatrix} \quad c = \begin{bmatrix} 3 \\ -3 \\ 3 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$B = b - A \cdot \frac{A^T . b}{A^T . A}$$

$$= \begin{bmatrix} 2 \\ 0 \\ -2 \end{bmatrix} - \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \frac{[1 \ -1 \ 0] \begin{bmatrix} 2 \\ 0 \\ -2 \end{bmatrix}}{[1 \ -1 \ 0] \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}} \quad 2 \quad 2$$

$$= \begin{bmatrix} 2 \\ 0 \\ -2 \end{bmatrix} - \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} \checkmark$$

$$C = c - \frac{A^T . c}{A^T . A} . A - \frac{B^T . c}{B^T . B} . B$$

$$= \begin{bmatrix} 3 \\ -3 \\ 3 \end{bmatrix} - \frac{[1 \ -1 \ 0] \begin{bmatrix} 3 \\ -3 \\ 3 \end{bmatrix}}{[1 \ -1 \ 0] \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}} . \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} - \frac{[1 \ 1 \ -2] \begin{bmatrix} 3 \\ -3 \\ 3 \end{bmatrix}}{[1 \ 1 \ -2] \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}} . \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$$

$$= \begin{bmatrix} 3 \\ -3 \\ 3 \end{bmatrix} - \frac{6}{2} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} - (-1) \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\therefore A = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \quad B = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} \quad C = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad \left| \begin{array}{l} A^T.B = 0 \\ B^T.C = 0 \\ A^T.C = 0 \end{array} \right.$$

$$\therefore q_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \quad q_2 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} \quad q_3 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$